CSC 242 Artificial Intelligence (Spring 2026)

Key Information

Course Description

This course introduces fundamental principles and techniques from Artificial Intelligence, including heuristic search, automated reasoning, handling uncertainty, and machine learning in order to prepare students for advanced AI courses.

Credit Hours

This is a four credit class. This class follows the university credit hour policy and is expected to to require a typical time commitment of 8-12 hours per week.

Meeting Time and Location

TR 450pm-605pm in Wegmans Hall 1400

Prerequisites

CSC 172, MATH 150, and CSC 173 strongly recommended.

Textbook

Artificial Intelligence: A Modern Approach, Russel and Norvig, 4th edition.

Instructor

- Instructor: Adam Purtee
- Homepage: cs.rochester.edu/u/apurtee.
- Email: apurtee AT cs DOT rochester.edu
- Office: Wegmans Hall 2109

Key Information Instructor Assignments Exams Grading Schedule Policies

Assignments

There will be five programming assignments this semester, corresponding to the overall modules of search, representation, uncertainty, and learning. The programming assignments will be graded using a combination of automated test cases and human TA review. Each project will require you to implement a specific algorithm (or algorithms) from the course, to perform some testing and experiments, and to share your results in a clear, well-written report. All projects in this course are individual assignments (i.e., no group work).

A small percentage of your grades will come from *Engagement*, which is meant to incentivize active learning and course participation, including responding to in-class activities and short surveys.

Exams

There are three exams in this course: two midterms, and a final. Your lowest midterm will count for 10%, your highest midterm for 20%, and the final exam will count for 20% for a total of 50% of your overall grade.

All exams will be traditional, in-person, paper quizzes. No notes or electronics are permitted - smartphones, smartwatches, and headphones are all prohibited. Students found using electronic devices will be reported to the Board of Academic Honesty.

Students with documented accessibility issues should plan to communicate in advance with the disability services office to arrange proctoring. Students with academic conflicts with quizzes (such as away games or performances) should contact the instructor as soon as possible to work out a solution.

Students must present their university ID when turning in exams.

Grading

Your Overall Numeric Grade

Your scores on the individual components of this course will be weighted to obtain your course score. All appeals of grades on individual components must be made within one week of the grade being available. The following table represents the weighting of course components.

Exams	50%
Projects	45%
Engagement	5%
Total	*100%*

Letter Grades

Letter grades will follow the Official University of Rochester Grading Scheme . Note that the University scheme puts “average” somewhere between C and B. The following table is an estimate of how numeric grades map onto letter grades.

Letter Grade	**Threshold**
A (Excellent)	≥ 94%
A–	≥ 90%
B+	≥ 87%
B (Above Average)	≥ 84%
B–	≥ 80%
C+	≥ 77%
C	≥ 74%
C– (Minimum satisfactory)	≥ 70%
D (Minimum passing)	≥ 60%
E	< 60%

Schedule

Date	**DoW**	**Topic**	**Reading**	**Assignments**
Jan 20	T	Introduction	1.3-1.5	

Jan 22	R	Problem Solving	2.1, 2.3, 2.4, 3.1-3.4	Project 1 assigned; due Feb 9.
Jan 27	T	Informed Search	3.5, 3.6	
Jan 29	R	Local Search	4.1-4.4	
Feb 3	T	Case Study: Beam Decoding		
Feb 5	R	Adversarial Search	5.1-5.2	Project 2 assigned; due Feb 23.
Feb 10	T	Pruning and Sampling	5.3-5.4	
Feb 12	R	Constraint Satisfaction Problems	6.1-6.5	
Feb 17	T	Propositional Logic	7.1-7.4, 7.5.1	
Feb 19	R	Satisfiability	Class Notes	Project 3 assigned; due Mar 2.
Feb 24	T	First Order Logic	8.1-8.2, 8.3	
Feb 26	R	First Order Reasoning	7.5, 7.6, 9.1, 9.2, 9.5	
Mar 3	T	Review		
Mar 5	R	Midterm 1		
Mar 10	T	No Class (Spring Break)		
Mar 12	R	No Class (Spring Break)		
Mar 17	T	Representing Uncertainty	12.1, 12.2, 12.3, 12.4, 12.5, 12.6	
Mar 19	R	Bayesian Networks	13.1, 13.2, 13.3	Project 4 assigned; due Apr 6.
Mar 24	T	Approximate Inference	13.4	

Mar 26	R	Sequential Models	14.1-14.5	
Mar 31	T	Learning From Examples	19.1-19.2, 19.4, 19.6	
Apr 2	R	Learning to Classify	19.6	Project 5 assigned; due Apr 27.
Apr 7	T	Neural Networks	21.1	
Apr 9	R	Backpropagation	21.2	
Apr 14	T	Review		
Apr 16	R	Midterm 2		
Apr 21	T	Deep Learning	21.3	
Apr 23	R	Reinforcement Learning	17.1-17.2, 22.1, 22.3	
Apr 28	T	Large Language Models	24.1-24.4	
Apr 30	R	The Future of AI	TBD	
May 6	W	Final Exam (715-915pm)		

Policies

Academic Honesty

All assignments and activities associated with this course must be performed in accordance with the University of Rochester's Academic Honesty Policy. More information is available at: www.rochester.edu/college/honesty

All assignments must be completed on your own. You may discuss the assignments at the level of ideas with your peers, but you should not directly exchange answers or source code.

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course - using AI to complete the projects is likely to undermine the learning process. Rather than adjudicate allowed vs disallowed cases, by default this course is permissive, with the requirement that you are transparent about your usage. Assignment writeups will require you to either briefly describe how you used AI, or to state that you did not use it. Any substantial use of AI without disclosure will be considered academically dishonest. The instructor may adjust this policy during the semester if necessary.

All incidents of academic dishonesty will be reported. **Violations of the academic honesty policy carry significant penalties, such as a zero on the assignment and additional reduction by whole letter grades. (I.e., from B+ to C+). Repeat offenders may be expelled from their majors and from the university.**

Disability Resources

The University of Rochester respects and welcomes students of all backgrounds and abilities. In the event you encounter any barrier(s) to full participation in this course due to the impact of disability, please contact the Office of Disability Resources. The access coordinators in the Office of Disability Resources can meet with you to discuss the barriers you are experiencing and explain the eligibility process for establishing academic accommodations. You can reach the Office of Disability Resources at: disability@rochester.edu; (585) 276-5075; Taylor Hall.

Students with an accommodation for any aspect of the course must make arrangements in advance through the Disability Resources office. Then, as instructed by the office, contact the instructor to confirm your arrangements.